

FRANCISCO JAVIER GUILLEN NAVARRO

Aerospace & Software Engineer | AI & Workflow Automation

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Professional Profile: Aerospace & Software Engineer with a solid background in space software, computer vision, and safety-critical real-time systems. Specialized in **internal workflow automation** and team efficiency optimization. Passionate about **AI-augmented development (Vibe Coding)**, leveraging advanced tools (Claude Code, OpenAI Codex, Antigravity) and REST APIs to rapidly prototype, build, and deploy full-stack mobile (Flutter) and web applications.

AI & TECH STACK

AI & Vibe Coding (Personal)

Claude Code

OpenAI Codex

Antigravity

REST APIs

Flutter / Dart

Web Prototyping

Core Engineering

Python

C/C++

Bash

MATLAB / Simulink

Git / SVN

Linux / Windows

Data Science & ML

- TensorFlow / Keras
- OpenCV (Computer Vision)
- NumPy, Pandas, SciPy
- Scikit-Learn, Matplotlib

EDUCATION

M.S. Aeronautical Engineering

Polytechnic Univ. of Madrid

Sep 2019 – Mar 2022

Spec: Space Vehicles.

Thesis: Heart disease detection via AI.

B.S. Aerospace Engineering

Polytechnic Univ. of Madrid

Sep 2015 – Jul 2019

Spec: Aerospace Propulsion.

Thesis: Hybrid Rocket Engine Design.

CERTIFICATES

Machine Learning Certificate

DeepLearning.ai

Neural Networks & Deep Learning

Stanford University

LANGUAGES

- Spanish:** Native
- English:** Intermediate

WORK EXPERIENCE

Space Software Engineer

Aug 2023 – Present

GMV • Madrid, Spain

- Developed safety-critical real-time aerospace software utilizing **Bash and Python**.
- Internal Workflow Automation:** Proactively designed and implemented custom automated Python/Bash tools to streamline team workflows and eliminate repetitive operational tasks.
- Excelled in high-pressure aerospace operations with strong teamwork.

Research Assistant

Jun 2023 – Present

Francisco de Vitoria University (BioCompLab) • Madrid, Spain

- Automated preprocessing of high-speed video and medical diagnostic test data using custom Python scripts.
- Provided technical computations for cellular fluid mechanics models.

Consultant Engineer

Apr 2022 – Aug 2023

Airbus DS (via Aertec Solutions) • Madrid, Spain

- Industrialized functional tests and automated test execution via MATLAB and Python scripting.
- Supported innovation initiatives with technical proposals and rapid solution prototyping.

Artificial Intelligence Researcher

Apr 2021 – Aug 2021

Kerox Technology • Madrid, Spain

- Researched computer vision algorithms, model selection, and deep learning training.

Systems Engineer (Grant)

Dec 2019 – Aug 2020

ISDEFE • Madrid, Spain

- Validated safety requirements and compliance for Air Traffic Management (ATM) systems.

FEATURED PERSONAL PROJECTS & AI

Cross-Platform Mobile App (Flutter & AI)

Vibe Coding • Claude Code • REST API Integration

- Architected and launched a full-stack Flutter mobile app using AI-driven workflows (Claude Code / Codex).
- Connected external REST APIs and backend tools to deliver dynamic real-time features.

AI-Driven Web Platforms & Interactive Tooling

Full-Stack Web • LLM APIs • Automated Workflows

- Developed web tools leveraging LLMs, prompt engineering, and modern agentic pipeline integration.